VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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MINI PROJECT

on

"SignBridge: A Real-Time Bidirectional Sign Language Communication System Using Continuous Gesture Recognition and AI-Driven Speech-to Sign Synthesis"

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I. Title of the Mini Project

"SignBridge: A Real-Time Bidirectional Sign Language Communication System Using Continuous Gesture Recognition and AI-Driven Speech-to-Sign Synthesis"

II. Abstract

Communication between sign language users and non-signers remains difficult in everyday settings such as hospitals, banks, colleges, and shops, since most existing tools depend on controlled backgrounds, good lighting, or user-specific calibration to recognize signs reliably. This project proposes SignBridge, a real-time hand-gesture recognition system built on computer vision and deep learning. A standard camera captures a live video feed of the signer's hand; a background-removal step isolates the hand from the surrounding scene during preprocessing, allowing gestures to be recognized reliably even under reduced lighting conditions. A CNN extracts spatial features from the isolated hand region while an LSTMbased sequence model captures temporal motion, allowing the system to recognize gestures directly from the live feed, without requiring images to be uploaded, and to work across different users despite variation in hand size, orientation, and appearance. Recognized gestures are converted into text and, optionally, synthesized speech, making the system directly useful as an accessibility tool. Unlike prior systems that rely on isolated signs, controlled backgrounds, or user-specific calibration, SignBridge unifies real-time, user-independent, hand-gesture-only recognition into a single camera-only application, aiming to make communication in healthcare, banking, education, and retail settings more accessible without dedicated sensors or a human interpreter.

III. Objectives

1. To collect a large and diverse image dataset of hand gesture signs, capturing variations across different users, hand orientations, backgrounds, and lighting conditions, to support robust and generalizable model training.
2. To train and validate a machine learning-based classification model using the collected static image dataset, extracting relevant hand-shape and feature descriptors for accurate real-time gesture recognition.
3. To integrate the trained recognition model into a functional, low-cost prototype application that converts recognized gestures into text and speech output, and to evaluate its recognition accuracy, response latency, and usability across normal and low-light conditions.

IV. Scope

The scope of SignBridge covers the collection of an image-based hand gesture dataset and the training of a machine learning-based model for static gesture recognition. The system processes individual image frames captured from a standard RGB webcam or smartphone camera in real time, recognizing discrete hand gestures without specialized sensors, gloves, or markers. It is intended for service-oriented environments such as hospitals, banks, colleges, and retail counters, where sensor-free gesture recognition can support communication between sign language users and non-signers. The current scope includes dataset collection across varied users and lighting conditions, background-removal preprocessing, feature extraction, frame-level gesture classification, and conversion of recognized gestures into text and optional speech output. The prototype will be evaluated for recognition accuracy, latency, and usability under normal and low-light conditions. Continuous sentence-level signing, video-based temporal modeling, bidirectional translation, and multi-dialect support remain outside the current scope and are addressed under Future Scope.

V. Societal Impact & Outcomes of the Project

This project addresses a communication barrier faced daily by the deaf and hard-of-hearing community in essential interactions such as healthcare, banking, education, and retail, where the absence of a shared language often causes delay, dependency on trained interpreters, or exclusion. An affordable, camera-only, two-way translation tool reduces this dependency, allows sign language users to communicate independently, and encourages more inclusive public and commercial spaces. Expected outcomes include a working real-time sign-to-speech and speech-to-sign prototype, measured accuracy and latency benchmarks, and a reusable gesture dataset that can later be expanded to additional vocabulary and sign languages. Beyond immediate use, the project supports the broader goal of accessible, assistive technology and can be extended into public-service kiosks, mobile applications, and tools for teaching sign language to hearing users

VI. References

[1] A. O. Hashi, S. Z. M. Hashim, and A. B. Asamah, "A systematic review of hand gesture recognition: An update from 2018 to 2024," IEEE Access, vol. 12, pp. 143603–143623, 2024, doi: 10.1109/ACCESS.2024.3421992.

[2] U. Rastogi, R. P. Mahapatra, and S. Kumar, "Advancements in Machine Learning Techniques for Hand Gesture-Based Sign Language Recognition: A Comprehensive Review," Archives of Computational Methods in Engineering, vol. 32, pp. 4265–4302, 2025, doi: 10.1007/s11831-025-10258-z.